KOTLIN MULTIPLATFORM

Building Nightly

The All-in-One Open-Source Music & Cinema Player



- Zero Ads & Tracking
- Android & Desktop
- In-App Cinema
- 28-Band FFT Audio
- AI Voice Search

A Complete Engineering Case Study & Showcase

github.com/aaweshdas/Audix-Multiplatform-Music-Player

THE PROBLEM

Why Are Modern Media Apps So Frustrating?

Invasive Ads & Clutter

Unskippable video and audio commercials interrupting playlists every 10 minutes.

Paywalled Basic Features

Background play, lyrics, offline downloads, and seeking locked behind subscriptions.

Fragmented Catalogs

Forced to juggle Spotify for music, YouTube for indie tracks, and piracy-riddled sites for cinema.

Resource Bloat & Telemetry

Heavy Electron wrappers eating 2GB+ RAM with aggressive background tracking.

The Solution:

Build an open-source, privacy-first client that does it all under one roof.

ARCHITECTURE

One Codebase. Native Performance.

ANDROID CLIENT

- AndroidX Media3 (ExoPlayer)
- Foreground Service Playback
- Lock Screen Notifications
- Hardware Media Key Hooks
- Dynamic Material 3 You Theming
- Low Battery & Memory Profile
- Min SDK 24 up to Android 15

DESKTOP CLIENT

- VLCJ Native libVLC C-Bindings
- Lossless Gapless Audio Engine
- Windows, macOS & Linux Support
- Multi-Window Video Playback
- 28-Band Real-Time Visualizer
- Standalone Portable Zero-Install
- Hardware GPU Acceleration

90%+ Shared Kotlin Code

State machines, SQLDelight database, network pipelines, and UI layouts are completely shared across

CAPABILITIES

Universal Music & In-App Cinema

Decoupled Extension SPI

Plug-and-play architecture pulling tracks directly from Spotify, YouTube Music, JioSaavn, SoundCloud, and RadioBrowser with automatic fallback mirrors.

In-App Cinema Theater

Embedded streaming studio with automated Cinemeta IMDb resolution (`tt...`), episode indexes, and multi-server stream mirrors.

Sandboxed Security Shield

Custom WebView interceptor blocking unwanted ad redirects, popups, and intrusive third-party scripts.

Synced Synchronized Lyrics

Real-time scrolling karaoke lyrics powered by LRCLIB, Genius, and Musixmatch.

DEEP TECH

Audiophile Visualizer & AI Voice

28-Band Cooley-Tukey Radix-2 FFT

- Mathematical Fast Fourier Transform computed over raw 512-point audio chunks.
- Hann windowing applied to eliminate edge spectral leakage.
- Logarithmic frequency scaling across 28 bands (20 Hz sub-bass to 20 kHz treble).
- Hardware-accelerated rendering at fluid 60fps.

Dual AI Speech-to-Text Engines

- Whisper Large V3 Turbo running via Groq for ultra-low-latency transcription.
- Deepgram Nova-2 fallback with real-time Voice Activity Detection (VAD).
- Hands-free voice querying for songs, artists, albums, and cinema movies.
- Zero audio storage or tracking — privacy by design.

GET STARTED

100% Free & Open-Source

Nightly is ready for you to explore, test, and contribute.

Check Out The Repository

https://github.com/aaweshdas/Audix-Multiplatform-Music-Player

Star the repo on GitHub if you love KMP & open source

Download the Android APK or Windows Portable Zip

Clone the project and run locally in Android Studio / IntelliJ

Connect with me on LinkedIn to discuss KMP & mobile tech!

Crafted with passion by Aawesh Das

Software Engineer & Cross-Platform Enthusiast

Tech Stack: Kotlin Multiplatform • Compose • ExoPlayer • libVLC • AI/STT

Drop a comment or share this with anyone exploring Kotlin Multiplatform!